

UNIVERSITI MALAYA  
UNIVERSITY OF MALAYA

PENILAIAN ALTERNATIF IJAZAH SARJANA MUDA SAINS KOMPUTER (KEJURUTERAAN  
PERISIAN)  
*ALTERNATIVE ASSESSMENT FOR THE DEGREE OF BACHELOR OF COMPUTER SCIENCE  
(SOFTWARE ENGINEERING)*

SESI AKADEMIK 2025/2026 : SEMESTER 2  
*ACADEMIC SESSION 2025/2026 : SEMESTER 2*

WIF3001 : Pengujian Perisian  
*Software Testing*

22 Jun 2026  
Masa: 24 jam

22 June 2026  
Time: 24 hours

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ARAHAN KEPADA CALON:  
*INSTRUCTIONS TO CANDIDATES:*

Calon dikehendaki menjawab semua soalan.  
*Answer all questions.*

(Kertas soalan ini mengandungi 1 soalan dalam 5 halaman yang dicetak)  
*(This question paper consists of 1 question on 5 printed pages)*

## **PROJECT BRIEF (WIF3001 – SOFTWARE TESTING)**

**University/Programme/Course:** University of Malaya/Bachelor of Computer Science (Software Engineering) /Software Testing

**Year:** 3<sup>rd</sup> year / 6<sup>th</sup> Semester

**Pedagogical Approach:** Project-based Learning

**Mark:** 50%

### **Learning Outcomes:**

Experience a basic design project:

- a. Download and execute the chosen open source software
- b. Elaborate the functionality of the software by listing the software requirements
- c. Analyze the structural and dynamic behavior of the software's program code
- d. Formulate the Software Test Plan using black box approach
- e. Design the tests based on the strategies formulated
- f. Execute the test cases with test data based on the Software Test Plan
- g. Apply ethical principles, standards and professional computer ethics
- h. Function effectively as a team member
- i. Check the coverage of each test conducted
- j. Communicate effectively on complex computer science activities through reports and presentation

### **Objective:**

This project requires you to design a complete test plan and conduct tests for an open source software to achieve full test coverage using variety of test techniques.

### **Scope:**

To meet the project requirements, you will need to:

- Form a work team of 5 members and organize the necessary activities you will need to do to complete the blueprint of the documentations. Note: it is important that your team identifies clear roles and responsibilities, distributing and coordinating various tasks appropriately, and able to operate as a high performing team. You may assign the workload based on requirements assigned on.
- Search for an open source web-based system from previous Final Year Projects (or from Github) that has a complete use case documentation and other software design diagrams. You must make sure the software can be executed well in your machine. You have to study the functionality of the software by reading the manual (if available), the use case documentation, or execute the software several times to gather the software requirements. You must test all requirements (or use cases) of the system. Each member must have equal number of requirements (or use cases) to test. Please write member's name for each requirement's test design and related section has been prepared.
- Design the tests using black box (compulsory techniques: Use Case Testing, GUI Testing, Questionnaire for UAT and scenario testing) approach. You can refer to the system's state diagrams to develop test cases using scenario testing approach. The project will use at least 10 techniques (including Use Case Testing and GUI Testing). The chosen techniques will

determine on how much coverage of the program code you will conduct. You must have justifications for each technique chosen. One technique can be used by many requirements.

- Use software metrics tools to measure the quality of the program code. You can use TestRail and SonarCube to check the project. If you use other technologies, please report it in the documentation.
- Choose five requirements where you can put test script (for test automation) to ensure the program code maintains their behaviors as expected to be.
- Propose new requirements (including non functional requirements) to improve the project. You have to redesign the use case diagram based on your proposed requirements. You must explain and justify your proposed requirements.
- You must report on all AI technologies have been used to deliver the test plan.
- Present your test results and project work experiences via an oral presentation using a recorded video of 10 minutes. All members must present. Your presentation must clearly communicate how you:
  - Worked as a team (e.g., challenges faced and how they were tackled, etc)
  - Key decisions made (what basis and how these are decided) concerning the design
  - Technologies used
  - Test Automation strategy
  - Proposed requirements
  - Results of the tests

## **Learning Guides:**

Week 8: Project is distributed to the students via Spectrum. Students have to form a group of 5 members and assigned role. They have to use Group Contract form that is available in the Spectrum.

Week 9: Submit the Group Contract

Week 9: Discuss and decide one open source software to be tested among team members.

Week 8: Submission and feedback on chosen project

Week 10: Discussion on progress (upon request)

Week 15: Submit a report that consists of

1. Test item transmittal report
2. Test plan
3. Test design specification (with justification)
4. Test case specification (please make sure there is no redundancy)
5. Test Incident Report
6. Test Summary Report
7. Measure the project quality
8. Test scripts for automation
9. New requirements proposal
10. Technologies used.

Week 15: Submit video recording

**Assessment:**

The distribution of marks for the test blueprint in Week 13 is contained in **Table 1**.

Table 1: Distribution of Marks

| Assessment Components                                                                                                                                                                                                                                                                                                                                             | Mark (%)    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Design<br>1. Test item transmittal report (5%)<br>2. Test plan (10%)<br>3. Test design specification (15%)<br>4. Test case specification (10%)<br>5. Test Incident Report (5%)<br>6. Test Summary Report (5%)<br>7. Measure the project quality (5%)<br>8. Test scripts for automation (10%)<br>9. New requirements proposal (10%)<br>10. Technologies used. (5%) | 80%         |
| Teamwork<br>• Group contract (GC) (5%)                                                                                                                                                                                                                                                                                                                            | 5%          |
| Communication<br>• Technical Writing (properly done – with name labelled) (5%)<br>• Oral Presentation (5%)<br>• Graphical Representations (5%)                                                                                                                                                                                                                    | 15%         |
| <b>Total</b>                                                                                                                                                                                                                                                                                                                                                      | <b>100%</b> |

**Submission Guideline:**

- 1- Only Group Leader has to do the submission
- 2- You must use this file name format for all documents –  
WIF3001-P-LeaderShortName\_Document.fileFormat  
For example: WIF3001-P-Hafizah\_GC.pdf
- 3- You must submit two version of file formats (docx/pptx and pdf) for each document you submit except for Group Contract (pdf only)

These are the documents you will submit:

1. Group Contract (GC)
  2. Software Brief Description (SBD) – the original report written by the developer
  3. Report (Report)
  4. PowerPoint Slide (PPT)
- 4- On Week 15 – 22<sup>nd</sup> June, 2026 at 12.00pm,
    - a. Please submit via Spectrum.
    - b. Please zip GC, SBD, ppt, video, and the report.

THE END